VIGNAN'S LARA
INSTITUTE OF TECHNOLOGY & SCIENCE
Approved by AICTE New Delhi & Affiliated to JNTUK Kakinada
An ISO 9001 : 2015 Certified Institution

INDUSTRIAL INTERNSHIP REPORT

(Smart Home Automation using Embedded Systems and IoT)

Prepared by
Polisetty Srivalli
B.Tech – 3rd Year (ECE)

Vignan's Lara Institute of Technology and Science
Vadlamudi, Guntur, Andhra Pradesh

Submitted to

Upskill Campus
In collaboration with
UniConverge Technologies Pvt Ltd

EXECUTIVE SUMMARY

This report provides details of the Industrial Internship conducted through **upskill Campus** and **UniConverge Technologies Pvt Ltd**.

The internship focused on developing a **Smart Home Automation System using Embedded Systems and IoT** technologies. The project was completed within six weeks.

The main objective of this project was to design and implement a system that allows users to control home appliances remotely using a microcontroller and communication modules.

This internship helped in gaining practical knowledge about embedded systems, IoT technologies, and real-world engineering solutions.

TABLE OF CONTENTS

1. Preface
2. Introduction
    2.1 About UniConverge Technologies Pvt Ltd
    2.2 About Upskill Campus
    2.3 Objectives
    2.4 References
    2.5 Glossary
3. Problem Statement
4. Existing and Proposed Solution
5. Proposed Design / Model
6. Performance Test
7. My Learnings
8. Future Work Scope

1. PREFACE

This report summarizes the work completed during the six-week internship program. Industrial internships play an important role in bridging the gap between academic learning and practical industry applications.

During this internship, I worked on the project **Smart Home Automation using Embedded Systems and IoT**. The project aims to automate home appliances such as lights and fans using wireless communication.

The internship program was organized by **Upskill Campus** in collaboration with **UniConverge Technologies Pvt Ltd**.

This internship provided valuable experience in system design, programming, and project documentation.

---

2. INTRODUCTION

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd is a technology company established in 2013. It focuses on digital transformation and industrial solutions using modern technologies such as:

- Internet of Things (IoT)
- Machine Learning
- Cloud Computing
- Cyber Security
- Communication Technologies

The company provides solutions in smart factories, predictive maintenance, and industrial automation.

2.2 About Upskill Campus

upskill Campus is a career development platform that provides internships and training programs to improve students' technical skills and employability.

It connects students with industry experts and real-world projects.

2.3 Objectives of the Internship

The objectives of this internship were:

- To gain practical industry experience
- To understand real-world engineering problems
- To improve technical knowledge
- To develop problem-solving skills
- To implement IoT and embedded systems projects

Block Diagram – Smart Home Automation

```
+-------------+       +--------------------+       +------------------+
|   Mobile    | -----> |  WiFi / Bluetooth  | -----> |     Arduino      |
| Application |       |      Module        |       |  Microcontroller |
+-------------+       +--------------------+       +------------------+
                                                        |
                                                        |
                                                        v
                                                  +---------------+
                                                  |  Relay Module |
                                                  +---------------+
                                                        |
                                                        |
                                                        v
                                                  +---------------+
```

```
| Home Appliances|
| (Light / Fan)  |
+---------------+
```

Code of project

```
// Smart Home Automation using Arduino and Bluetooth

char data;              // variable to store data
int light = 7;          // light connected to pin 7
int fan = 8;            // fan connected to pin 8

void setup()
{
  Serial.begin(9600);

  pinMode(light, OUTPUT);
  pinMode(fan, OUTPUT);

  digitalWrite(light, LOW);
  digitalWrite(fan, LOW);
```

```
}

void loop()
{
  if (Serial.available() > 0)
  {
    data = Serial.read();
  }

  // Light Control
  if (data == 'A')
  {
    digitalWrite(light, HIGH);
  }

  if (data == 'a')
  {
    digitalWrite(light, LOW);
  }

  // Fan Control
  if (data == 'B')
  {
    digitalWrite(fan, HIGH);
  }

  if (data == 'b')
  {
```

```
    digitalWrite(fan, LOW);
  }
}
```

Components Used

- Arduino Uno
- HC-05 Bluetooth Module / ESP8266 WiFi Module
- Relay Module
- Light / Fan (Appliances)
- Jumper Wires
- Power Supply

---

3. PROBLEM STATEMENT

Traditional home appliances require manual operation, which can be inconvenient and inefficient.

The goal of this project is to develop a **Smart Home Automation System** that allows users to control appliances remotely using IoT technology.

The system uses a microcontroller and communication modules to enable wireless control of devices.

---

4. EXISTING AND PROPOSED SOLUTION

Existing System

Traditional home automation systems are limited and require manual control.

Limitations include:

- Lack of remote access
- Limited automation
- Higher operational effort

Proposed Solution

The proposed system uses:

- Arduino Microcontroller
- IoT communication module
- Relay module for appliance control

This allows users to control appliances through wireless communication.

---

5. PROPOSED DESIGN / MODEL

High Level Diagram

User → Mobile Application → Communication Module → Arduino → Relay → Appliance

Low Level Components

- Arduino Uno
- Bluetooth / WiFi Module
- Relay Module
- Power Supply
- Household Appliances

---

6. PERFORMANCE TEST

The system was tested for functionality and response time.

Test Plan

- Check device connectivity
- Test switching operations
- Measure response time

Performance Outcome

The system successfully controlled appliances with minimal delay.

---

7. MY LEARNINGS

Through this internship, I learned:

- Embedded systems programming

- IoT communication methods
- Hardware interfacing
- Debugging and troubleshooting
- Technical documentation

This internship improved my technical skills and industry knowledge.

---

8. FUTURE WORK SCOPE

The project can be further improved by adding:

- Mobile application control
- Cloud-based monitoring
- Voice assistant integration
- Energy consumption monitoring
- Smart security systems